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StoryNode: a node-based visual orchestration framework for automated comic and storyboard generation

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Abstract
Maintaining visual and narrative consistency across sequential frames remains a critical bottleneck in digital storytelling. This paper presents StoryNode, an AI-assisted multimedia orchestration framework designed to streamline comic and storyboard generation for novice visual storytellers. By formalizing visual generation as a Directed Acyclic Graph (DAG) orchestration problem, StoryNode integrates Large Language Models (LLMs) for narrative parsing with a context-aware diffusion pipeline and a human-in-the-loop refinement layer. The framework makes three primary contributions: (1) Context-Driven Consistency: Context-Driven Consistency: a fine-tuning-free feature-anchoring workflow that preserves character and style continuity across sequential panels without additional per-character optimization or user-side model-weight updates; (2) End-to-End Workflow Integration: a pipeline connecting LLM scriptwriting, panel generation, style adaptation, and speech-bubble composition; and (3) Accessible Workflow Orchestration: a node-based interface that abstracts complex diffusion parameters into task-oriented controls. We evaluated StoryNode in a counterbalanced within-subjects user study (N=42) examining what novice users could accomplish with StoryNode and a technically strong IP-Adapter-Plus–ControlNet baseline after the same standardized 10-minute onboarding. The study therefore evaluates short-onboarding accessibility and workflow friction rather than the expert-level capability ceiling of either system. Under this controlled onboarding condition, average production time decreased from 8.7 to 3.4 h, task success increased from 38.8% to 90.5%, and standardized usability measures favored StoryNode (SUS: 82.4; NASA-TLX: 34). These findings suggest that DAG-based orchestration can reduce operational friction and support human-in-the-loop visual storytelling with existing generative models.

Keywords Generative AI · Multimodal orchestration · Directed acyclic graph · Character consistency · Human-in-the-loop · Digital storytelling

1 Introduction

Comics and storyboards are sequential visual media in which images, text, layout, and panel-to-panel transitions jointly carry narrative meaning. Recent generative models can produce high-quality single images, yet creating a coherent visual sequence remains difficult for users without professional illustration or AI-workflow experience. In practice, novice creators must repeatedly manage prompts, reference images, poses, masks, style settings, and text overlays across many panels. The central bottleneck has therefore shifted from generating an isolated image to orchestrating a multi-stage, multimodal workflow that maintains visual continuity while preserving creative control.

To address this workflow-level challenge, this paper presents StoryNode, an orchestration framework built on

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ComfyUI, DeepSeek-R1 [42], and FLUX.1 Kontext [43]. StoryNode treats ComfyUI not merely as a graphical interface, but as a Directed Acyclic Graph (DAG) orchestration engine in which narrative parsing, context-aware image generation, stylization, and speech-bubble composition are connected as explicit modules. We explicitly note that StoryNode does not claim a brand-new model-level diffusion architecture. While contemporary training-free models such as StoryDiffusion [33] and Story2Board [35] optimize internal attention mechanisms for story image generation, StoryNode targets a system-level orchestration challenge: reducing workflow friction, enabling interactive human-in-the-loop adjustment, and automating multimodal typesetting around existing generative models.

The target users are independent game developers, multimedia prosumers, students, and small creative teams who have story ideas but limited formal illustration training. In StoryNode, users provide a core story idea, the LLM generates structured storyboard prompts, the diffusion workflow produces and refines panels, and users intervene through human-in-the-loop nodes when narrative or visual intent requires adjustment. This framing positions StoryNode as an accessibility and orchestration contribution, not as a replacement for professional artists or as evidence that generative AI alone can solve visual storytelling.

Compared with conventional manual diffusion workflows, StoryNode aims to reduce production friction while preserving points of human control. The main contributions of this paper are:

- Modular Node-Based Orchestration: We propose a visual workflow architecture that decouples script reasoning, image generation, style adaptation, and text composition into controllable micro-services within a DAG.
- End-to-End Visual Storytelling Pipeline: We design and implement a pipeline that connects LLM narrative parsing, context-aware panel synthesis, multi-style conversion, and speech-bubble integration in a single executable workflow.
- Progressive Human-in-the-Loop Control: We introduce a refinement mechanism in which automated drafting is paired with user-directed node adjustment, and we evaluate whether this design reduces cognitive load and workflow attrition for novice creators.

2 Background and related work

StoryNode is situated at the intersection of sequential visual storytelling, diffusion-based character consistency, and human-in-the-loop workflow design. This section narrows the discussion to work that directly informs the paper's system contribution: how comics combine text and images, how recent generative systems address story visualization, and why node-based orchestration matters for novice users.

2.1 Sequential visual storytelling and comics

Comics are not simply collections of images; they are sequential compositions in which panels, gutters, captions, speech balloons, and page layout jointly structure narrative meaning. Classic semiotic accounts distinguish anchorage, where text constrains the interpretation of an image, from relay, where text and image advance the narrative together [31]. Comics scholarship further emphasizes that meaning emerges across panel transitions, spatial arrangement, and the reader's reconstruction of time between images [1, 6]. These properties make comic generation different from single-image generation: the system must preserve identity and style while supporting panel rhythm, text placement, and local revision.

For this reason, StoryNode treats visual storytelling as an orchestration problem. A useful system must coordinate narrative planning, visual consistency, composition, and typographic integration across multiple frames, while still allowing the creator to intervene when the generated sequence fails to match intent.

AI-Assisted Story and Comic Generation.

Prior work on story visualization and comic generation has explored several complementary directions. Make-A-Story introduces visual memory for consistent story generation [32]. StoryDiffusion improves long-range consistency through consistent self-attention [33]. ConsiStory addresses training-free subject consistency across prompts [34], and Story2Board targets expressive storyboard generation with latent panel anchoring and attention-based feature mixing [35]. These systems are important direct context for StoryNode. However, most focus primarily on model-level consistency or story-to-image generation; StoryNode instead contributes a workflow layer that integrates existing models with editable DAG routing, speech-bubble composition, and human-in-the-loop refinement. Table 1 summarizes these systems in relation to training-free generation, sequential narrative support, character consistency, text integration, and workflow-level HITL support.

2.2 Character consistency in diffusion-based generation

Diffusion-based consistency workflows commonly combine spatial control, identity reference, and fine-tuning. ControlNet provides conditional spatial guidance such as pose, edge, and depth maps [36], while IP-Adapter introduces image-prompt conditioning that can preserve identity

Table 1 Method Capability Comparison for Sequential Visual Storytelling

Method	Requires task-/character-specific fine-tuning	Sequential narrative	Character consistency	Speech-bubble/text integration	Workflow/ HITL support
Make-A-Story [32]	Yes	Yes	Visual memory	No	Limited
StoryDiffusion [33]	No	Yes	Consistent self-attention	No	Model-level generation
ConsiStory [34]	No	Prompt sequence	Shared attention / feature injection	No	Model-level generation
Story2Board [35]	No	Storyboard prompts	Latent panel anchoring	No	Limited workflow support
StoryNode	NO	Yes	Kontext anchoring + user refinement	Yes	ComfyUI DAG + HITL

features without full model fine-tuning [37]. LoRA and DreamBooth-style approaches can achieve strong personalization but require curated training images and additional optimization. These methods are technically powerful, but they often expose novice users to complex parameter choices, reference preparation, and repeated manual routing across panels.

StoryNode does not replace these underlying methods. Instead, it wraps the relevant capabilities into a task-oriented DAG so that narrative parsing, identity anchoring, spatial control, style adaptation, and local corrections can be executed and revised in one workflow. This distinction is central to the evaluation: the baseline is technically strong, but operationally demanding; StoryNode's contribution is to reduce the friction of using such capabilities for sequential storytelling.

2.3 Human-in-the-loop workflow orchestration

Generative AI can support non-professional creators by accelerating ideation and lowering the cost of visual iteration, but full automation can also weaken agency, make errors difficult to correct, and obscure the logic of the workflow [10, 15, 28]. For creative tasks, a more appropriate design goal is often not full autonomy, but progressive control: the system handles repetitive technical operations while the user retains authority over narrative intent, local revision, and final selection.

Node-based systems are well suited to this goal because they make intermediate operations visible and editable. StoryNode builds on this principle by exposing narrative, visual, and typographic stages as modular nodes. The resulting workflow is intended to help novice users move from prompt-level experimentation to structured visual direction without requiring professional drawing skills or deep ComfyUI expertise.

3 Formative research

We conducted a formative study with 42 novice visual storytellers recruited through online communities. Participants had no professional training in illustration or comics, but they expressed interest in using visual narratives for personal stories, entertainment, learning, or small creative projects. A pre-screening questionnaire excluded professional illustrators, art students, and experienced ComfyUI users so that the study reflected the target audience of StoryNode.

3.1 Participants and procedures

Participants ranged in age from 16 to 32 years, with a mean age of 24.5 years. The cohort included college students, software engineers, teachers, marketing personnel, healthcare workers, and freelancers. Four participants had previously tried image-generation tools such as Midjourney or Stable Diffusion through online platforms, but none reported experience with complex node-based workflows such as ComfyUI. Eleven participants had never used AI image-generation tools. All participants reported basic computer and Internet literacy.

Eight participants had made preliminary attempts at comic creation, such as drawing a single image, making simple doodle comics, or planning a short story visually. The remaining participants had story ideas but had not attempted visual production. Across the cohort, participants reported recurring difficulties in character design, style continuity, panel-to-panel narration, and production efficiency.

The formative study combined a questionnaire with semi-structured interviews. After providing informed consent, participants completed a demographic and background questionnaire covering age, occupation, education, prior visual-creation experience, AI-tool experience, and target use cases. Interviews were conducted online and lasted approximately one hour. Participants described previous attempts at comic or AI-assisted visual creation, identified

pain points, and discussed what forms of control they would expect from an AI-assisted workflow.

3.2 Thematic analysis and design requirements

We coded the interview data thematically and grouped the most frequent barriers into four categories: character identity drift, anatomical and spatial distortion, style inconsistency, and composition control failure. Table 2 summarizes the distribution of these bottlenecks across the 42 participants. The analysis was used to derive system requirements rather than to make broad claims about all AI-painting users.

The formative results led to four design requirements for StoryNode. First, the workflow must preserve character and style continuity across panels. Second, it must provide controllable spatial and compositional constraints without requiring users to manually build every ControlNet or reference branch. Third, it must reduce workflow fragmentation by integrating narrative planning, visual generation, and speech-bubble composition. Fourth, it must preserve creative agency by allowing local human intervention rather than forcing a fully automatic pipeline.

3.3 Design requirements from novice pain points

Participants were interested in AI-assisted comic creation, but their expectations were constrained by concrete production problems rather than by abstract enthusiasm for generative AI. The following requirements guided the StoryNode design:

Table 2 Thematic Clustering of Primary Technical Bottlenecks ($N=42$)

Primary Technical Bottleneck Category	Count (N)	Percentage	Representative User Feedback (Excerpt)
Character Identity Drift (Inability to maintain facial/clothing features across frames)	16	38.1%	"The protagonist's hair color and glasses disappear or change randomly in the next panel." (P8)
Anatomical & Spatial Distortion (Limb deformation, perspective failure in multi-character scenes)	12	28.6%	"The generated actions violate basic physical constraints, like arms passing through solid objects." (P9)
Style Inconsistency (Fluctuations in shading, line-art, or ambient lighting)	8	19.0%	"It looks like three different artists drew the sequence; the ambient lighting constantly shifts." (P10)
Composition Control Failure (Inability to accurately direct camera angles or panel rhythm)	6	14.3%	"I ask for a top-down view of a fight, but it keeps generating flat, front-facing portraits." (P14)

R1. Visual consistency: Participants treated stable character identity and style as prerequisites for sequential storytelling. If the protagonist's face, clothing, or color palette changed from panel to panel, participants described the output as unusable regardless of individual image quality.

R2. Controllability: Participants wanted visible control over poses, panel composition, and local corrections. They were willing to accept AI-generated drafts, but they needed a way to adjust failures without restarting the entire sequence.

R3. Workflow continuity: Existing AI workflows were perceived as fragmented because users had to move repeatedly between prompts, reference images, masks, and post-editing tools. This fragmentation increased learning cost and task abandonment.

R4. Creative agency: Participants worried that excessive automation would make the work feel generic. This finding motivated the human-in-the-loop nodes that preserve user choice at key points in the visual narrative pipeline.

4 Principle of workflow design

To derive the system requirements, we combined findings from the formative study, inspection of existing ComfyUI-based creator workflows, and iterative prototyping across multiple comic themes. These iterations exposed recurring breakdowns in prompt decomposition, character preservation, style transfer, and post-generation text composition. The resulting design requirements were therefore to make the comic-generation process visually traceable, controllable, consistent across panels, and modular enough for local human intervention. These requirements also preserve user agency by allowing creators to guide the overall narrative and visual direction rather than delegating the entire process to automation.

R1. LLM-Driven Script and Storyboard Generation: The narrative foundation is constructed using DeepSeek-R1 [42]. Based on a foundational user prompt, the LLM outputs a structured sequence of storyboard descriptions, including scene context, shot type, character action, and draft dialogue. This converts an informal story idea into panel-level prompts that can be routed through the visual generation pipeline.

R2. Context-Aware Visual Synthesis (Text-to-Image): The structural scripts are iteratively processed by the Kontext text-to-image pipeline. Leveraging the model's sequential comprehension capabilities, prompt continuity is maintained dynamically. To mitigate inherent feature drift across generation iterations, the system incorporates a dual-image referencing mechanism. This allows the network to anchor newly generated frames against targeted reference images or specific stylistic prompt qualifiers (e.g., line-art

rendering), ensuring visual uniformity across the narrative sequence.

R3. Iterative Consistency Refinement and Multimodal Integration: To finalize the comic sequence, localized modifications and textual overlays are applied. The framework utilizes a single-image editing workflow to embed dynamic speech bubbles and integrate dialogue text directly into the generated panels. Furthermore, to correct structural anomalies or character inconsistencies common in standard diffusion models, a mask-based dual-image reference workflow is deployed. This mechanism autonomously aligns localized masked regions with a designated character reference, facilitating precise, non-destructive modifications.

Finally, we integrated R1–R3 into ComfyUI to implement an end-to-end AI-assisted comic-generation workflow.

5 StoryNode based on FLUX.1 kontext

Figure 1 summarizes the StoryNode workflow as four connected stages: structured script generation, context-aware panel synthesis, style adaptation, and speech-bubble composition with human-in-the-loop revision. The diagram marks the main modules, data inputs, and outputs used in the end-to-end comic-generation workflow. Figure 2 further details the hierarchical DAG architecture used to organize these modules.

5.1 Story generation via LLM

In this module, we utilized DeepSeek-R1 to autonomously generate structured narrative scripts. We prompted the LLM with a core thematic concept: exploring the emotional depth between humans and robotic entities. The model was instructed to output a 20-panel storyboard script, explicitly defining the visual scene, character actions, and accompanying text/dialogue for each frame. Table 3 presents a representative excerpt of this generated sequence (themed "Life is measured by connections, not circuits"), showcasing various challenging shot types including establishing shots, action sequences, and emotional close-ups. This structured textual output serves as the precise input data for the subsequent Context-Aware Visual Synthesis module [22].

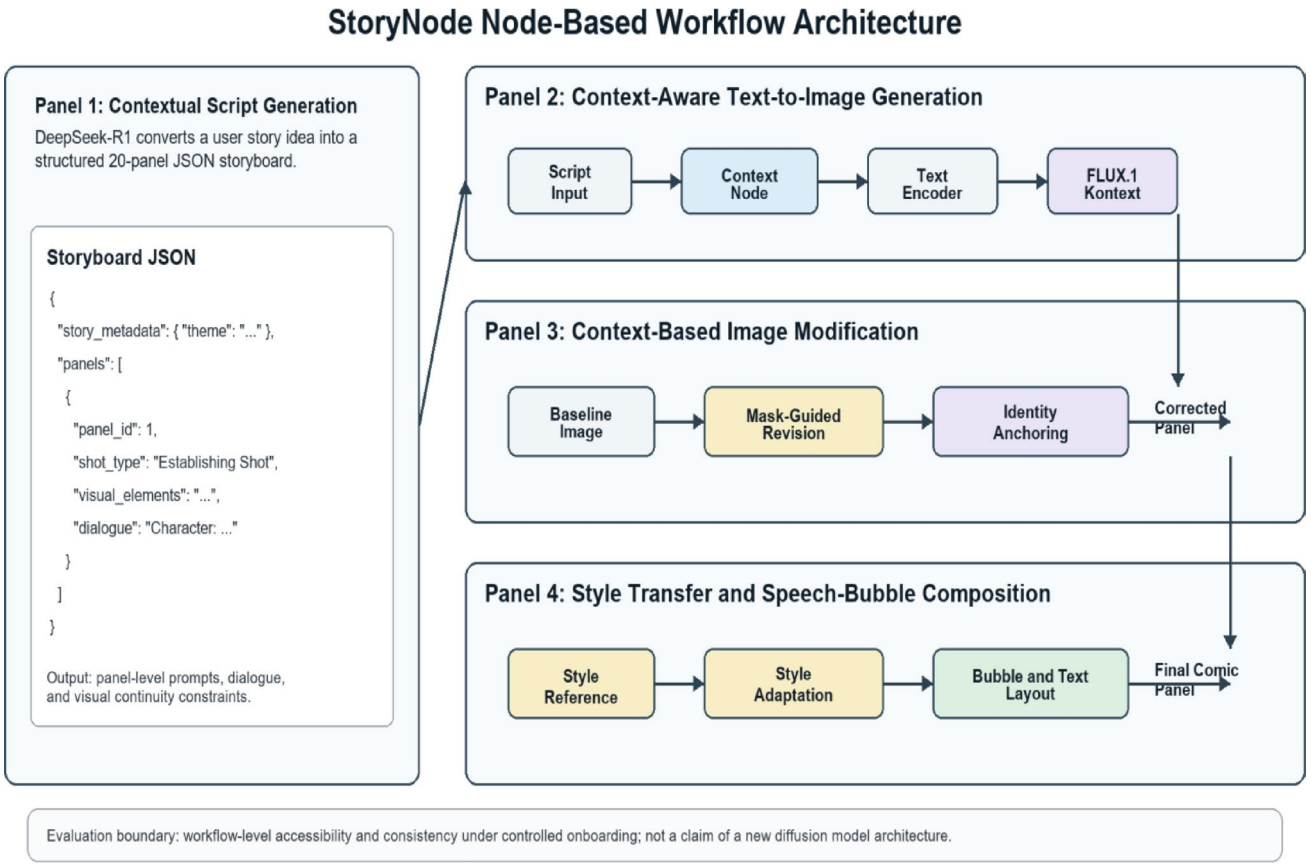


Fig. 1 StoryNode Node-Based Workflow Architecture for Comic and Storyboard Generation

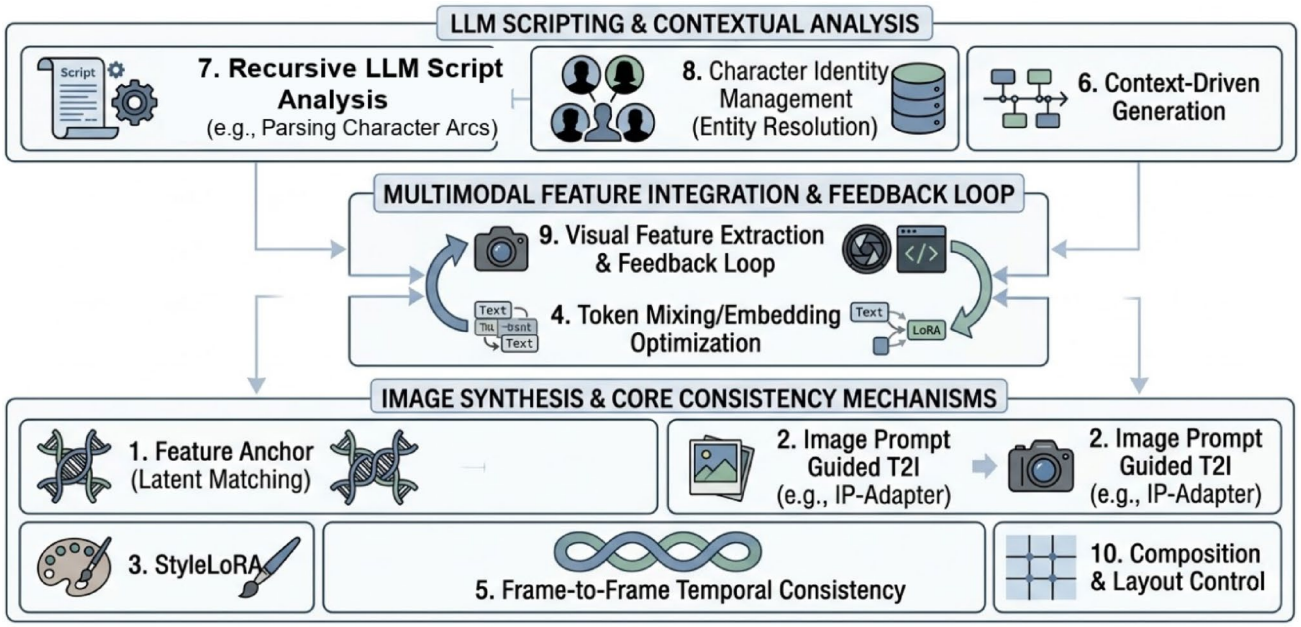


Fig. 2 Hierarchical DAG Architecture for Multimedia Orchestration and Consistency Management

Table 3 Representative Excerpt of LLM-Generated Structured Storyboard

Panel (Excerpt)	Shot Type / Narrative Phase	Visual Elements (Prompt Input for Kontext)	Dialogue / Narrative Text
Frame 1	Establishing Shot (World-building)	Rainy futuristic city skyline.	"New Haven, 2147. Humans and robots coexist… but not always connect."
Frame 5	Close-up (Detail & Repair)	Close-up of Arthur's hands soldering circuits. Helper Bot watches blankly.	"Late nights. Steady hands. Frayed wires."
Frame 11	Character Interaction	Helper Bot points at the robot dog, questioning the logic.	"QUERY: WHY MAINTAIN DEFECTIVE UNIT?"
Frame 14	Action / Conflict	Robot dog malfunctions, knocking over a priceless vase.	"Shattered history."
Frame 20	Final Shot (Resolution)	Robot dog and Helper Bot walk into the city together, guiding a lost child.	"New bonds grow. Life continues."

5.2 Context-aware visual synthesis (Text-to-Image Pipeline)

Following the generation of the structured narrative script by the LLM, the parsed data is routed to the visual synthesis module. The framework leverages FLUX.1 [dev] and FLUX.1 Kontext [43] as the foundational text-to-image and in-context editing engines. StoryNode uses these models as external generative components and focuses on how their capabilities can be organized, constrained, and revised within a continuous DAG workflow. The automated text-to-image synthesis pipeline is shown in Fig. 3.

5.3 Stylization and feature alignment

To adapt the initial photorealistic outputs into a cohesive comic aesthetic, the workflow incorporates a fine-tuning-free style adaptation mechanism using the pretrained FLUX.1 Kontext model. Here, "fine-tuning-free" indicates that users do not train additional LoRA modules, optimize character-specific weights, or update the underlying model. Instead, the system performs in-context image editing and text-conditioned transformation while keeping the pretrained model weights fixed (Fig. 4). The resulting sketch representation emulates traditional black-and-white comic shading and provides a high-contrast basis for integrating typographic elements, such as speech bubbles, and applying downstream colorization [24, 30].

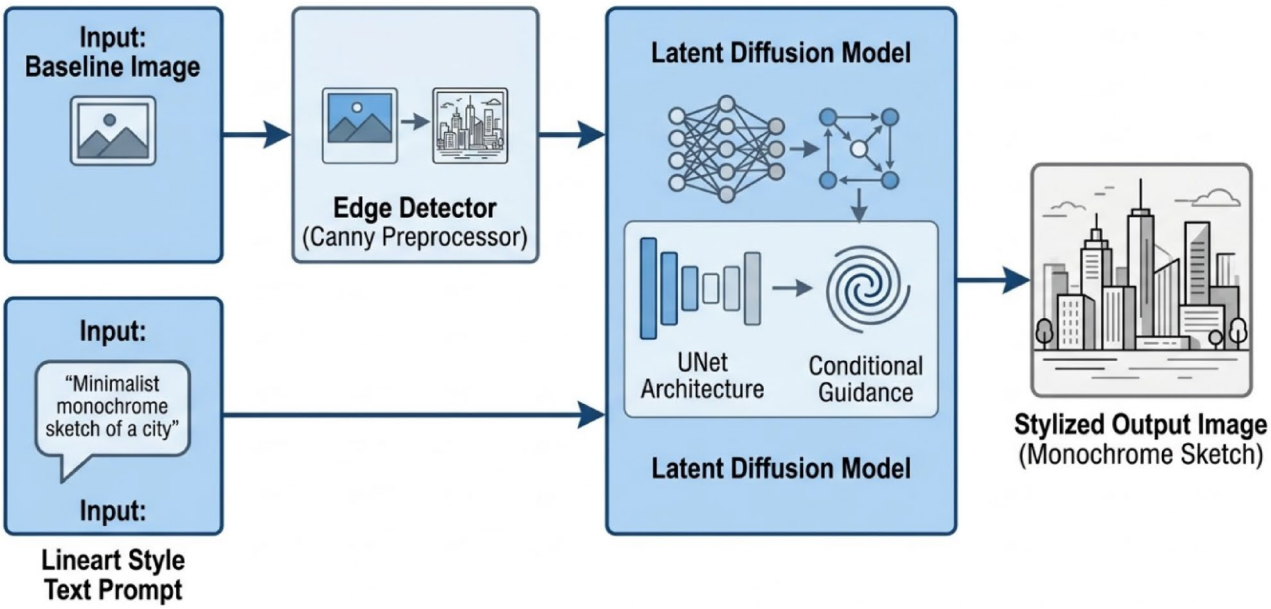


Fig. 3 Sequential Text-to-Image Synthesis Pipeline within the Automated Orchestration Layer

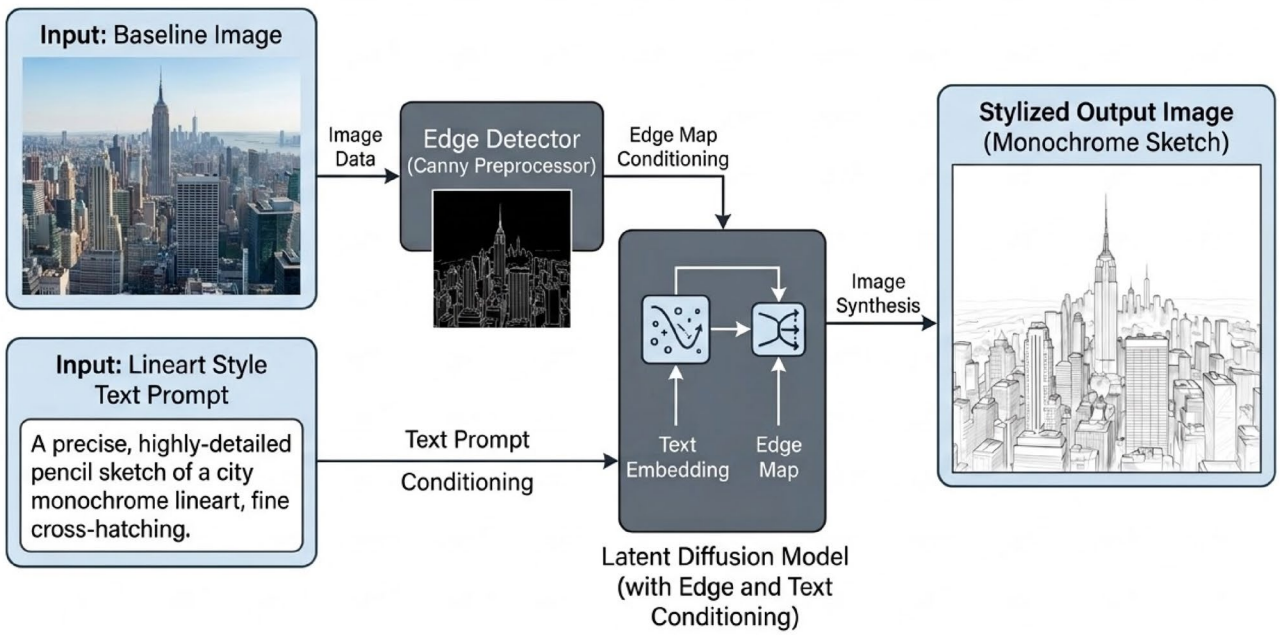


Fig. 4 Fine-Tuning-Free Style-to-Sketch Conversion through In-Context Editing

5.4 Localized modification and multimodal integration

The subsequent phase involves localized visual editing and typographical integration. To simulate authentic comic dialogue, the workflow employs the FLUX.1 Kontext architecture to dynamically embed localized text overlays (e.g., speech bubbles) directly into the generated panels (Fig. 5). Given the inherent limitations of diffusion models in rendering precise typography—often resulting in typographic hallucinations or garbled artifacts—the system supports a human-in-the-loop iterative refinement cycle.

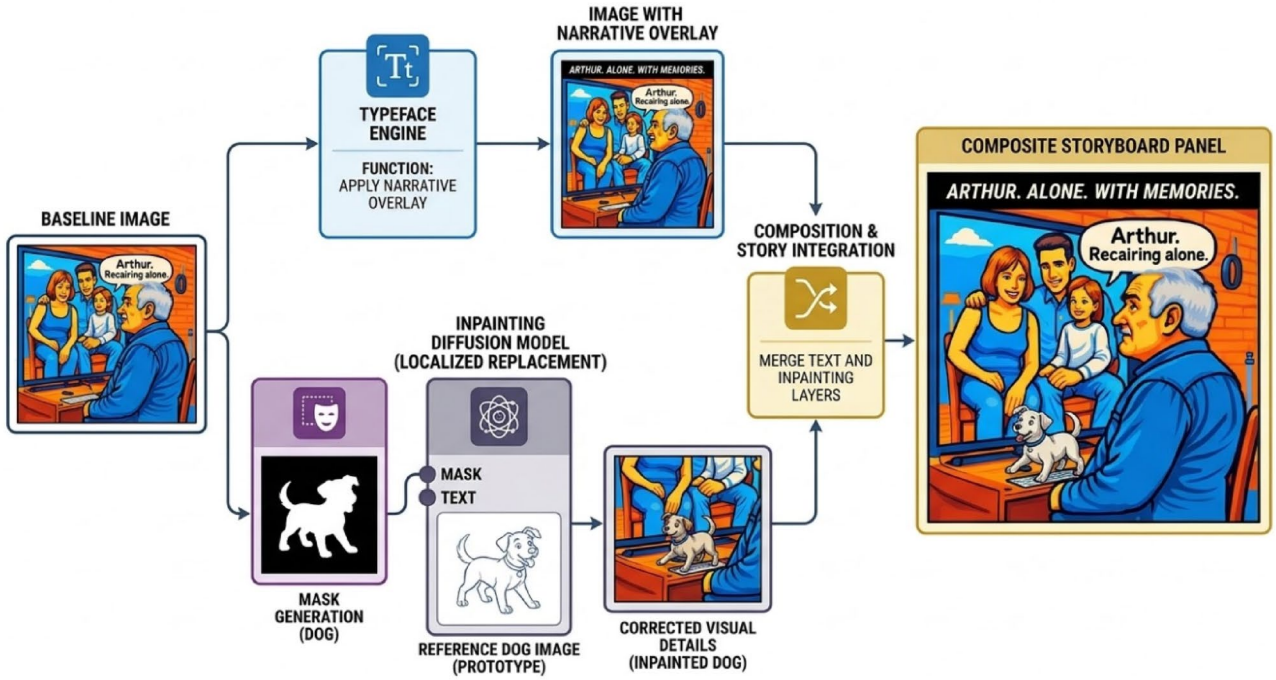


Fig. 5 Multi-Conditional Global Stylization and Domain Adaptation Pipeline

Furthermore, to resolve localized character identity drift (e.g., anatomical inconsistencies in secondary characters like the robot dog), the framework utilizes a mask-guided dual-image referencing mechanism (Fig. 5). This allows the user to selectively anchor specific masked regions to a reference prototype, ensuring strict morphological consistency without altering the global composition [20].

To resolve the tension between automated generation and precise creative intent, StoryNode facilitates a bifurcated refinement architecture (Fig. 5). This module allows users to selectively anchor character identities through a mask-guided dual-image referencing mechanism, while simultaneously leveraging a typographic engine for localized text embedding. By decoupling the visual repair from the narrative overlay, the framework ensures that speech bubbles and character consistency remain unified within a high-fidelity composite panel.

5.5 Global style transfer and domain adaptation

To support diverse creative intents, StoryNode incorporates a flexible global stylization module capable of domain adaptation, allowing for seamless transitions between diverse aesthetics such as American comic, Japanese manga, or impressionistic styles (as illustrated in Fig. 7). The underlying global stylization workflow (detailed in Fig. 6) enables two distinct modes of style migration: (1) Image-prompted transfer, which anchors the generation to the visual aesthetic of a provided reference image, and (2) Text-driven stylization, which modulates the output based on semantic style qualifiers. Leveraging the Kontext engine's temporal awareness, the system preserves the underlying semantic context of the scene across sequential frames, ensuring that the applied stylization remains visually coherent throughout the narrative sequence [16].

6 System architecture

At the user-interaction layer, the ComfyUI front end functions as a visual workflow engine that coordinates the back-end services [17]. Users provide an initial story description, and node operations are mapped to micro-service calls that return generation progress and intermediate outputs in real time. At the scenario-generation layer, StoryNode uses DeepSeek-R1 through its official API and a structured prompt template to produce panel-level storyboard data. At the visual-consistency layer, the core mechanism is fine-tuning-free, user-side feature anchoring driven by FLUX.1 Kontext.

This terminology refers only to the absence of additional task- or character-specific training and does not imply that FLUX.1 Kontext itself is an untrained model. Unlike workflows that require additional LoRA fine-tuning for each character, StoryNode injects reference-image features into the generation process to maintain a persistent visual context

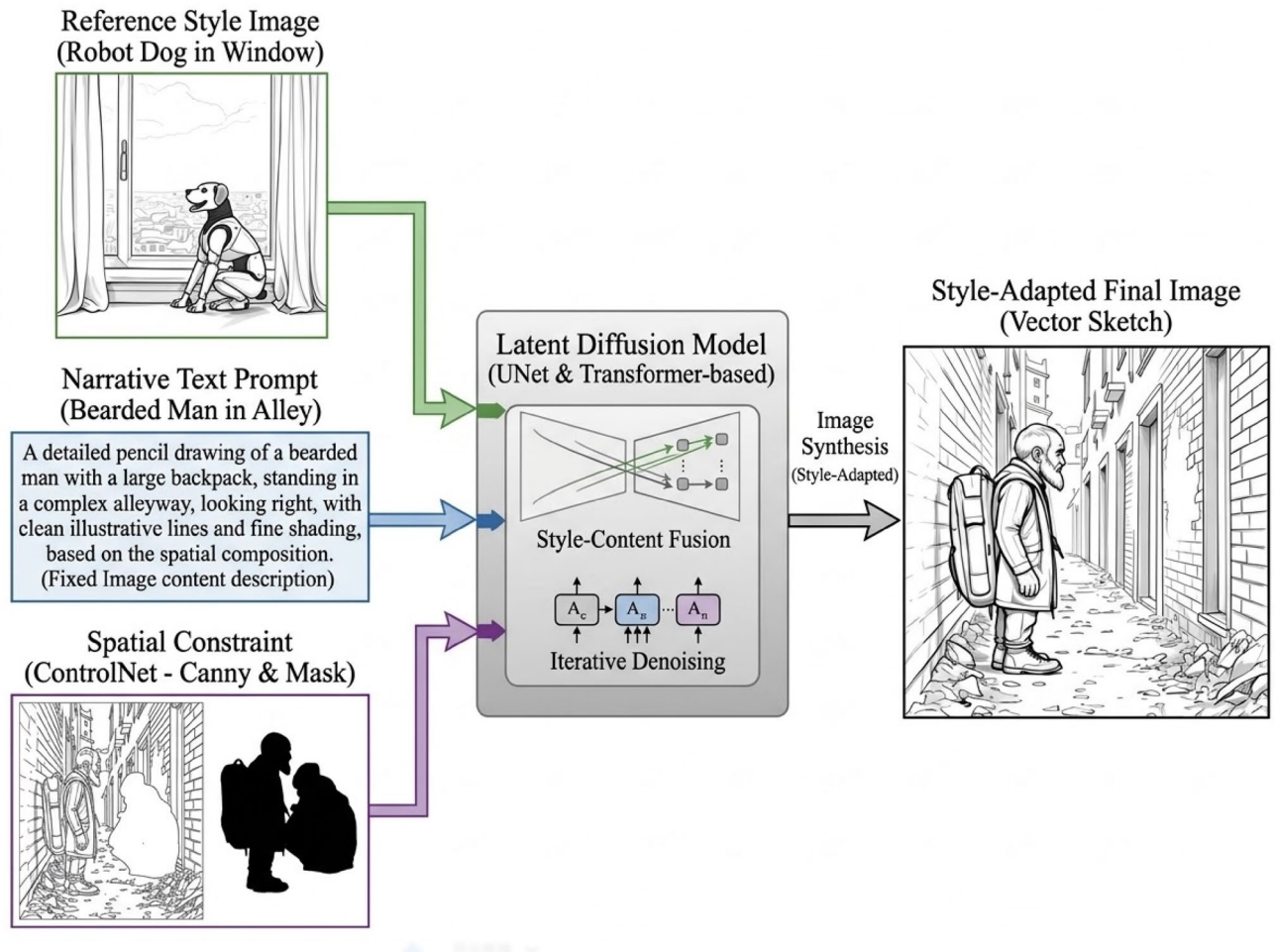


Fig. 6 Integrated Architecture for Localized Feature Refinement and Multimodal Narrative Composition

Table 4 System Architecture and Data Optimization Pipeline

Pipeline Module Interface	Data Modality & Format	Compression / Optimization Strategy
LLM to ComfyUI Orchestrator	JSON-serialized storyboard scripts	Dictionary compression (Zstandard)
Kontext Engine to Diffusion Model	Latent space embeddings (512-dimensional)	FP16 precision quantization
Diffusion Model to Style Transfer	RGB continuous bitmap (3-channel)	Lossy compression (WebP)
Text/Bubble Engine to Final Render	RGBA continuous bitmap (4-channel)	Block transmission coding

across sequential panels. By separating spatial layout guidance from semantic feature anchoring, the system supports temporal consistency without updating model weights. The system architecture, data transmission, and optimization strategies are summarized in Table 4.

Unlike fragmented workflows that suffer from repeated VRAM allocation overhead, StoryNode's unified DAG routing optimizes memory residency for preprocessors, contributing to the observed efficiency gains in long-horizon tasks.

7 Example use case

We used the StoryNode workflow to create and assemble a 20-panel comic sequence based on the human-machine story scenario described above [14]. The sequence demonstrates the system's main functions, including context-aware panel synthesis, style-transfer settings, and speech-bubble composition, as shown in Figs. 7 and 8.

8 Summative user evaluation

We conducted a summative within-subjects study to evaluate whether StoryNode reduces workflow friction for novice visual storytellers. The evaluation focused on end-to-end accessibility and task completion rather than isolated model quality. Participants completed the same 20-frame narrative



Fig. 7 Comparative Visualization of Cross-Domain Stylization and Domain Adaptation

task under two workflow conditions: StoryNode and a High-Standard Baseline.

The High-Standard Baseline combined IP-Adapter-Plus for identity preservation with ControlNet for spatial and pose constraints [36, 37]. This baseline represents a technically strong ComfyUI-based workflow for character-consistent generation, but it requires manual node routing, reference preparation, parameter tuning, and repeated correction across frames. StoryNode uses comparable underlying generative capabilities but abstracts them into a novice-oriented DAG with integrated speech-bubble composition and human-in-the-loop revision.

The comparison was intentionally designed as a short-onboarding accessibility evaluation. It does not estimate the maximum performance achievable by trained practitioners using IP-Adapter-Plus and ControlNet. Rather, it asks how effectively novices can operate each workflow after an identical, limited introduction. Before each condition, participants received a standardized 10-minute tutorial on the

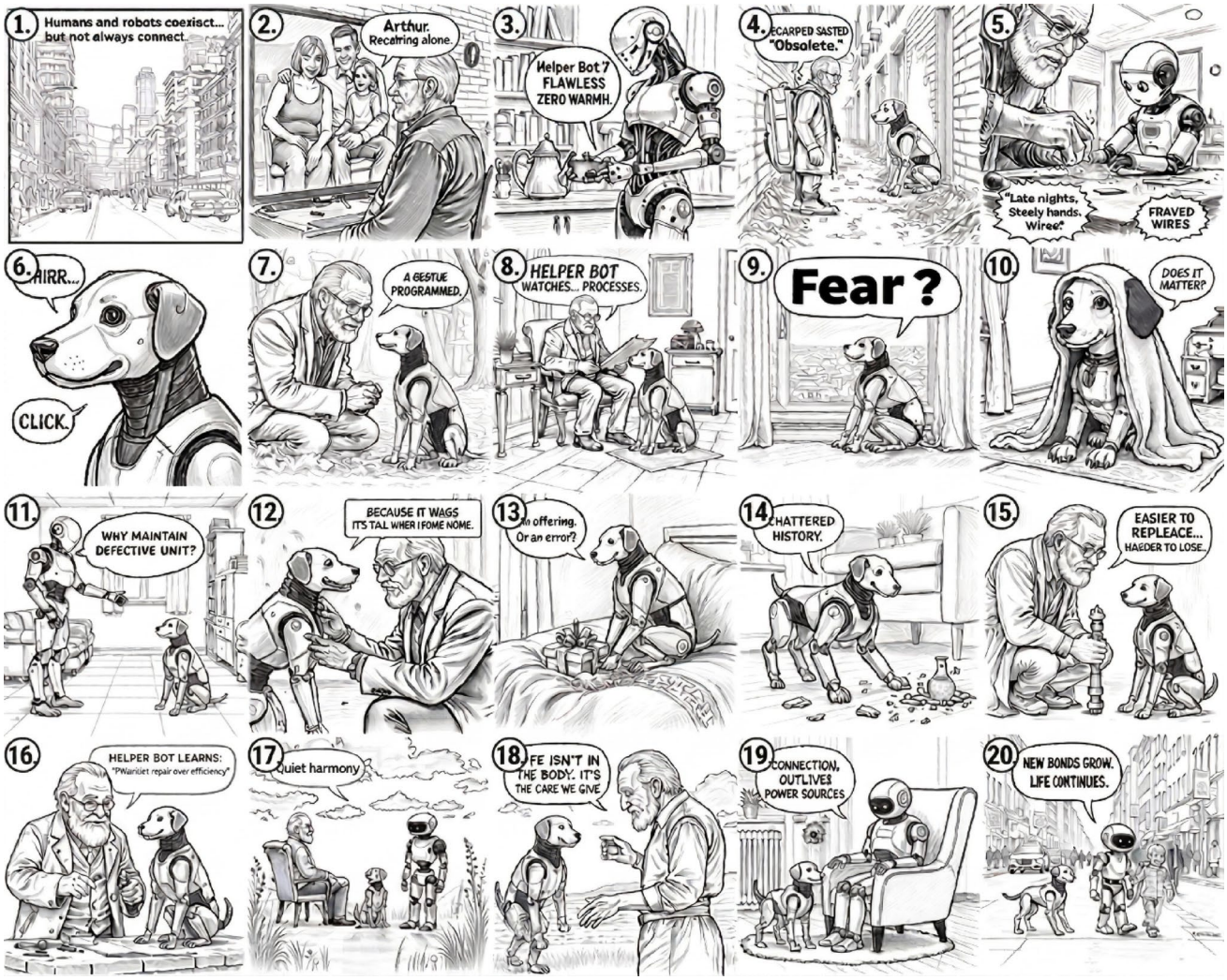


Fig. 8 Representative 20-Frame Narrative Sequence Generated via the StoryNode Framework

relevant workflow. They were then asked to create a short comic sequence depicting a character's growth through adversity. The task required character continuity, cross-panel narrative coherence, and at least one speech-bubble or text-overlay operation. After each condition, participants completed questionnaires and participated in a short debriefing interview.

8.1 Participants and procedures

For the summative evaluation, we recruited a separate cohort of 42 inexperienced participants ($N=42$) who had not taken part in the formative interviews. The formative study and summative study used two independent, non-overlapping cohorts of 42 participants each. This separation was intended to reduce priming from the requirement-gathering phase and to provide an independent usability evaluation of the completed workflow. Cohort non-overlap was verified before analysis using anonymized participant identifiers and separate recruitment records.

We used a two-condition within-subjects design. Each participant completed both the StoryNode condition and the High-Standard Baseline condition, using the same foundational diffusion model, target output length, task prompt, and evaluation criteria. The comparison therefore measures workflow-level accessibility and efficiency under controlled task requirements.

The task asked users to design a short comic featuring a character's growth after experiencing setbacks. This theme was intentionally open enough to support individual narrative variation while still requiring character consistency, scene-to-scene continuity, and text-image integration.

To reduce order effects and fatigue, system exposure was counterbalanced. Participants were randomly divided into two equal order groups:

- Group A ($n=21$) completed the High-Standard Baseline first and StoryNode second.
- Group B ($n=21$) completed StoryNode first and the High-Standard Baseline second.

Table 5 Average Latency per Pipeline Stage (Per Panel)

Pipeline Stage	Average Latency (Seconds)	Percentage of Total Machine Time
LLM Script Parsing & Prompt Routing	1.2 s	5.3%
Context-Aware Visual Synthesis (FLUX)	14.5 s	63.6%
Fine-Tuning-Free Feature Anchoring	4.3 s	18.9%
Style Transfer & Dynamic Bubble Integration	2.8 s	12.2%
Total Automated Pipeline Latency	**22.8 s**	**100%**

After each condition, we collected task completion, completion time, subjective consistency ratings, SUS, NASA-TLX, and interview feedback. The debriefing focused on where participants experienced workflow friction, whether they felt able to recover from generation errors, and whether the system preserved enough creative control.

To support reproducibility without publicly releasing the full database before publication, we prepared a reproducibility package containing the model variants, prompt templates, parameter settings, random-seed policy, image resolution, ComfyUI workflow JSON files, and anonymized analysis scripts. These materials can be provided to editors or reviewers during peer review when required and will be released in a public repository after publication. The full hyperparameter and system-configuration matrix is provided after the reference list as Appendix Table A1.

8.2 System performance evaluation

To validate the computational efficiency and operational viability of the StoryNode orchestration framework, we conducted a comprehensive system performance analysis. All benchmarks were executed on a standardized hardware environment equipped with an NVIDIA RTX 4080 GPU (16GB VRAM) and 64GB of system RAM. We evaluated the architecture across three critical system dimensions: Latency, Throughput, and Scalability.

8.2.1 System performance evaluation

We measured the execution latency of each decoupled micro-service within the Directed Acyclic Graph (DAG) pipeline. By breaking down the time consumption per stage for a standard single comic panel, we identified the primary computational bottlenecks. As shown in Table 5, the context-aware visual synthesis remains the most resource-intensive phase. However, the automated pipeline completely eliminates the human idle time typically spent manually routing data between these stages.

Throughput measures the system's efficiency in generating complete narrative sequences. We compared the continuous generation throughput of StoryNode against the High-Standard Baseline (IP-Adapter+ControlNet). Because StoryNode abstracts the workflow into an automated DAG, it bypasses the repetitive loading of preprocessors and manual VRAM clearing required in the baseline.

8.2.2 Throughput efficiency (macro-benchmark)

- Baseline Throughput: 1.2 frames per minute (heavily constrained by ControlNet pose extraction and manual node triggering).
- StoryNode Throughput: 2.6 frames per minute.

By orchestrating the modules into a unified continuous flow, StoryNode achieved a 116% increase in machine throughput in our benchmark setup. This result should be interpreted as a workflow and scheduling gain, not as evidence that the underlying diffusion model itself is faster.

A critical requirement for multimedia systems is the ability to scale to long-horizon tasks without performance degradation. We tested the system's scalability by generating comic sequences of varying lengths (10, 20, 30, and 50 frames) to observe the relationship between frame count and processing time.

8.2.3 Scalability (frame count vs. time)

We tested the system's scalability by generating comic sequences of varying lengths (10, 20, 30, and 50 frames) to observe the relationship between frame count and processing time.

The scalability analysis (Table 6) shows an approximately linear relationship between frame count and total generation time in the tested range. Long-horizon generation still produced minor VRAM caching overhead at 50 frames, so the result should be treated as evidence of practical scalability under the reported hardware conditions rather than a general guarantee for all devices or model variants.

The reported latency and throughput values should not be assumed to transfer directly to GPUs with 8GB VRAM. On such devices, the full 1024×1024 FLUX workflow may require sequential model loading, CPU offloading, reduced precision, lower-resolution generation, or disabling parallel branches. These adaptations are expected to increase latency and reduce throughput, although we did not benchmark a specific 8GB configuration. Cloud execution and lightweight or quantized model variants may provide practical alternatives for novice users without high-end local hardware.

Table 6 System Scalability (Frame Count vs. Generation Time)

Total Sequence Length	Total Generation Time	Average Time Per Frame	System Stability
10 Frames	3.8 min	22.8 s	Stable
20 Frames	7.6 min	22.8 s	Stable
30 Frames	11.5 min	23.0 s	Stable
50 Frames	19.8 min	23.7 s	Minor VRAM caching overhead

8.3 System architecture ablation study

To isolate the contribution of each core architectural component within the StoryNode framework, we conducted a system-level ablation study. We designed three degraded versions of the system by systematically removing key modules and evaluated these variants across 20-frame narrative generation tasks using objective visual consistency, measured by CLIP image similarity. This ablation is reported as an implementation analysis rather than as an additional multi-condition user study.

The experimental conditions were defined as follows:

- Full StoryNode: The complete proposed framework including the DAG orchestrator, Kontext anchoring, and Human-in-the-Loop (HITL) manual intervention.
- w/o Kontext (No Feature Anchoring): The fine-tuning-free feature anchoring module was bypassed. The system relied solely on text prompts and standard diffusion generation without dual-image attention.
- w/o DAG (Linear Pipeline): The non-linear Directed Acyclic Graph orchestration was replaced with a rigid, linear script-to-image pipeline. Parallel processing (e.g., simultaneous style transfer and speech bubble generation) was disabled.
- w/o HITL (Fully Automated): The human-in-the-loop nodes (such as manual mask-editing and parameter fine-tuning) were locked. The system forced a single-pass, fully automated generation based strictly on the LLM script.

To quantify workflow-level friction, we introduce Full-Panel Regeneration Attempts (FPRA) per 20-frame sequence as a behavioral proxy metric (Table 7). FPRA measures how often users must fully re-generate a panel due to unresolved structural or semantic issues that cannot be corrected via localized node-level editing.

As shown in Table 7, FPRA provides complementary insights beyond CLIP-based visual similarity. First, removing the Kontext-based feature anchoring is associated with a substantial increase in regeneration frequency (9.6 ± 2.2), indicates that identity drift is difficult to recover through local corrections alone. Second, disabling the DAG structure increases workflow inefficiency (4.5 ± 1.1), is consistent with that linearized pipelines reduce the system's ability to isolate and localize errors. Third, in the absence of HITL intervention, FPRA becomes 0 because no user-driven correction is permitted; this does not indicate improved performance, but instead reflects the removal of interactive

Table 7 Ablation Study Results on System Modularity

System Configuration	CLIP Similarity (Mean ± SD)	FPRA per 20-frame Sequence (Mean ± SD)	Local Recovery Support	Primary Failure Mode
Full StoryNode	0.842 ± 0.045	1.2 ± 0.4	Fully Supported	No major consistency failure observed
w/o Kontext	0.615 ± 0.073	9.6 ± 2.2	Supported but unstable	Severe character identity drift; local patches fail
w/o DAG	0.830 ± 0.050	4.5 ± 1.1	Mechanically Limited	Rigid routing; downstream errors require full re-run
w/o HITL	0.812 ± 0.061	0.0 (N/A)	Completely Disabled	No correction loop; artifacts persist

Table 8 Short-Onboarding Accessibility Comparison for Novice Users

Measure	High-Standard Baseline	StoryNode	Statistical test
Completion time	8.7 h	3.4 h	Paired t-test: $t(41) = 14.28$, $p < 0.001$, $d = 2.84$, 95% CI [4.54, 6.06] h
Task success rate	38.8%	90.5%	Exact McNemar: chi-square(1) = 19.12, $p < 0.001$
CLIP consistency	0.715	0.842	Descriptive consistency metric; no model-level claim
SUS	48.5	82.4	Paired t-test: $t(41) = -11.36$, $p < 0.001$
NASA-TLX	78	34	Paired t-test: $t(41) = 12.91$, $p < 0.001$

Note. Both conditions followed a standardized 10-minute tutorial. Results characterize novice accessibility under controlled onboarding and should not be interpreted as a comparison of expert-level workflow ceilings

correction capability. Meanwhile, visual artifacts persist in this condition (CLIP: 0.812 ± 0.061).

8.4 Baseline workflow comparison

To avoid conflating study conditions with untested model-level systems, the summative quantitative comparison reports only the two workflows that participants actually used: StoryNode and the High-Standard Baseline. Manual diffusion, standalone ControlNet, standalone IP-Adapter, StoryDiffusion, ConsiStory, and Story2Board are discussed as related methods or workflow context, but they are not reported as participant-tested conditions in this evaluation.

The comparison intentionally evaluates end-to-end accessibility rather than isolated model quality. The baseline is technically powerful because it combines IP-Adapter-Plus and ControlNet, but it is operationally demanding for novice users. StoryNode abstracts the same class of operations into a DAG workflow with explicit narrative, visual, and typographic stages.

1. Table 8 summarizes the two-condition results. The results should be interpreted as workflow-level gains: StoryNode reduced manual node routing, repeated parameter switching, and cross-tool operations, which improved task completion and reduced perceived workload for novice participants.
2. Because each participant completed both conditions, completion time, SUS, NASA-TLX, CLIP consistency, and subjective ratings were treated as paired measurements. Continuous outcomes were assigned paired-samples tests after normality screening, with Wilcoxon signed-rank tests reserved for non-normal paired distributions. Binary task completion was analyzed using an exact McNemar test.
3. A paired-samples t-test revealed that StoryNode significantly reduced average production time from 8.7 h to 3.4 h ($t(41) = 14.28$, $p < 0.001$, Cohen's $d = 2.84$, 95% CI [4.54, 6.06] hours). Standardized usability also improved: SUS increased from 48.5 to 82.4 ($t(41) = -11.36$, $p < 0.001$), while perceived cognitive workload measured by NASA-TLX decreased from 78 to 34 ($t(41) = 12.91$, $p < 0.001$). For binary task success, an exact McNemar test confirmed a significant increase from 38.8% in the High-Standard Baseline to 90.5% with StoryNode (chi-square(1) = 19.12, $p < 0.001$). These results support a workflow-level claim: StoryNode makes a technically strong but operationally demanding baseline substantially more usable for novice visual storytellers.

Table 9 Usability and Efficiency Comparison: StoryNode vs. High-Standard Baseline

Evaluation Dimension	Quantitative Evidence ($N=42$)	Qualitative User Feedback Corroboration
Skill barrier reduction	Task success increased from 38.8% in the High-Standard Baseline to 90.5% with StoryNode; abandonment decreased from 61.2% to 9.5%.	"I was intimidated by comic drawing before; now I can orchestrate it independently." (P3)
Production efficiency	Average completion time decreased from 8.7 h in the baseline to 3.4 h with StoryNode.	"Following a brief tutorial, my first cohesive frame was rendered." (P2)
Core value proposition	StoryNode reduced consistency anxiety by combining Kontext anchoring, DAG routing, speech-bubble integration, and HITL revision.	Participants emphasized that local correction and visible workflow stages made failures easier to recover from.

The strongest interpretation of Table 8 is therefore not that StoryNode exceeds every model-level system, but that it reduced workflow friction for novice users under controlled onboarding. The asymmetry in operational difficulty is treated as a study boundary: the High-Standard Baseline is technically strong but difficult for novices to operate after a short tutorial, while StoryNode evaluates whether orchestration can make comparable generative components more accessible.

8.5 Objective visual consistency evaluation

To complement the subjective user questionnaires, we used CLIP image-to-image similarity as an objective descriptive measure of cross-panel visual consistency within the tested workflows. We analyzed 840 generated panels, comprising 20 frames per participant across 42 participants.

Character and Style Consistency: We used CLIP image-to-image similarity to estimate temporal consistency of characters and styles across sequential frames [38]. Higher CLIP scores indicate stronger preservation of visual identity. StoryNode achieved an average CLIP similarity score of 0.842, compared with 0.715 for the baseline workflow. We do not interpret this value as evidence that StoryNode surpasses model-level systems such as StoryDiffusion [33] or Story2Board [35]. The high consistency score primarily reflects the capabilities of the underlying FLUX.1 Kontext engine, while StoryNode's contribution lies in organizing that capability into an editable DAG workflow with HITL revision and automated text-image composition [39].

Table 10 Subjective User Ratings on Consistency Metrics (Scale: 1–5)

Index	Mean ± SD	95% CI
Role consistency	4.50 ± 0.60	[4.31, 4.69]
Style consistency	4.30 ± 0.70	[4.08, 4.52]
Narrative coherence	3.90 ± 0.82	[3.64, 4.16]

Table 11 System Satisfaction and Controllability Feedback

Dimension	Mean ± SD	95% CI
SUS	82.4 ± 10.8	[79.0, 85.8]
Controllability	77.0 ± 12.4	[73.1, 80.9]
Satisfaction	88.0 ± 9.4	[85.1, 90.9]

8.6 Subjective usability and HCI metrics analysis

1. Does StoryNode mitigate the skill barrier for non-artists (as shown in Table 9)?

The quantitative and qualitative results suggest a substantial reduction in the creation threshold for inexperienced users. Compared with the High-Standard Baseline, which had a 61.2% abandonment rate due to workflow friction, StoryNode's abandonment rate was 9.5%. Users completed their first full sequential narrative in an average of 3.4 h, indicating a more accessible production pipeline for the target novice audience.

2. Subjective performance of style, role, and narrative consistency (as shown in Table 10).

Complementing the objective CLIP similarity results, the subjective ratings reveal a clear gap between visual consistency and narrative coherence. StoryNode achieved relatively high ratings for role consistency (4.50 ± 0.60) and style consistency (4.30 ± 0.70), whereas narrative coherence was lower (3.90 ± 0.82). This result suggests that StoryNode is more effective at preserving character identity and visual style than at guaranteeing causal, temporal, and emotional continuity across panels. Narrative coherence remains dependent on explicit transition planning, particularly for rapid scene changes, action–reaction sequences, and emotionally complex interactions. The underlying failure patterns and corresponding design implications are discussed further in Sect. 9.1.

3. Usability, controllability, and standardized system evaluation (as shown in Table 11).

To quantify user satisfaction and cognitive friction, we used two standardized Human-Computer Interaction scales: the System Usability Scale (SUS) for overall ease of use [40], and the NASA Task Load Index (NASA-TLX) for perceived workload [41].

When evaluating the proposed framework against the High-Standard Baseline, the differences in system friction were substantial. The baseline workflow, while technically capable of maintaining consistency, required manual orchestration of weights, spatial preprocessors, and multi-node

routing. This resulted in a high workload score (NASA-TLX: 78/100) and a marginal usability score (SUS: 48.5).

By contrast, StoryNode's node-based automation shielded novice users from many low-level parameter configurations. Participants reported an 'Excellent' usability score (SUS: 82.4) and lower cognitive load (NASA-TLX: 34/100).

These standardized HCI metrics support the qualitative finding that StoryNode reduces operational friction while preserving human-in-the-loop control. Some advanced parameter configurations in ComfyUI still confused absolute novices, but the overall node visualization helped participants understand and revise the workflow rather than abandoning the task.

9 Discussion

In the dimension of technology integration, the primary challenge is abstracting complex algorithms, such as the Kontext engine, into an accessible interface for novice users. The system design must reduce cognitive load through preset operational scenarios. By encapsulating complex parameters into task-specific preset nodes (e.g., an "action optimization package"), the framework translates complex attention mechanisms into intuitive user controls.

Regarding the interaction paradigm, while the node-based operation of ComfyUI offers high flexibility, it introduces a learning curve. Future iterations will benefit from hierarchical interaction designs, offering both a basic mode for one-click storyboard generation and an expert mode for granular parameter control [15, 18].

Furthermore, the allocation of control remains critical. Based on user research, non-professional creators prefer an orchestration balance heavily favoring automation but retaining essential manual intervention points. Crucial manual controls must be preserved for emotional expression calibration, narrative branching, and ethical safety verifications. Ultimately, the StoryNode framework demonstrates substantial potential in multimedia pipelines, specifically for rapid storyboard prototyping and generating visual novel assets.

9.1 Visual consistency does not guarantee narrative coherence

The lower narrative-coherence rating (3.90 ± 0.82), compared with role consistency (4.50 ± 0.60) and style consistency (4.30 ± 0.70), reveals an important distinction between preserving visual identity and preserving sequential meaning. A post-hoc inspection of the lowest-rated narrative sequences indicated that coherence was most vulnerable in three situations: action-to-consequence transitions in which the result of an action was not visually explicit; abrupt changes in location or time without an establishing or bridging panel; and multi-character interactions in which gaze direction, dialogue order, or emotional response did not clearly preserve turn-taking. In these cases, Kontext anchoring could maintain appearance while failing to represent causal, temporal, or emotional continuity. StoryNode therefore reduces visual drift more effectively than narrative-state drift.

To address this limitation, future versions will introduce a narrative-state node that explicitly tracks character goals, location, props, emotional states, and unresolved causal dependencies across panels. A transition-validation module will compare adjacent storyboard states before image generation and flag missing actions, abrupt spatial changes, or inconsistent emotional reactions. The system may then revise the prompt or insert a bridging panel before the sequence is rendered.

9.2 The progressive orchestration framework for multimodal AI

Beyond the specific implementation of comic generation, StoryNode exemplifies a broader design framework for generative AI tools, which we term the "Progressive Orchestration Paradigm." This framework addresses the inherent tension between automation (efficiency) and controllability (agency) through three transferable design principles:

Functional Decoupling via DAGs: The results suggest that complex generative tasks can benefit from being decomposed into a Directed Acyclic Graph (DAG) of independent micro-services. This allows users to intervene at specific nodes, such as narrative parsing or style transfer, without restarting the entire creative flow.

Abstraction of Latent Complexity: By encapsulating high-dimensional diffusion parameters into task-oriented nodes, the framework creates a more legible control layer for non-experts. This suggests a shift from pure prompt engineering toward structural orchestration in some creative AI tools.

Strategic Automation Balance: Our findings indicate that users did not want total automation. They preferred a system that handled repetitive technical labor while leaving semantic decisions, local corrections, and final selection to the human creator. StoryNode's observed success rate should therefore be interpreted as evidence for a balanced workflow design rather than for full automation.

9.3 Limitations

While StoryNode significantly mitigates the friction of digital storytelling, the current framework exhibits several intrinsic systemic boundaries. First, from a computational perspective, the context-aware anchoring mechanism experiences feature attenuation when processing extended narrative sequences (e.g., exceeding 50 continuous frames) due to compounding latent noise and stringent VRAM limitations. Second, in the semantic control domain, precise manipulation of physical camera parameters—such as extreme elevation angles or specific focal lengths—remains a challenge using purely text-driven prompts, frequently necessitating iterative generation cycles. Furthermore, the current style-transfer modality operates predominantly at the 2D surface texture level, struggling to autonomously replicate the nuanced spatial tension and deep emotional lighting characteristic of professional human authorship. Finally, from an HCI perspective, the node-based architecture still imposes a noticeable cognitive burden during granular parameter tuning and lacks resilient error-recovery mechanisms for accidental workflow disruptions. The reliance on high-end consumer GPUs (e.g., RTX 4080 with 16GB VRAM) also currently restricts the framework's broader accessibility, while the complex copyright attribution of multi-source synthesized assets remains an open industry challenge. Addressing these spatial control and VRAM optimization bottlenecks will be prioritized in future iterations [9].

While the 'growth through adversity' task successfully tested the system's ability to maintain narrative depth and character consistency, we acknowledge that its ecological validity is primarily centered on single-protagonist narratives. Future studies should introduce high-density multi-character interactions (e.g., ensemble casts) and complex architectural transitions to further stress-test the framework's spatial reasoning and feature anchoring limits.

A boundary of the efficiency comparison is the asymmetry in training difficulty. The High-Standard Baseline requires advanced knowledge of latent routing, preprocessor selection, and parameter weighting, so a short tutorial is insufficient for novices to achieve expert-level proficiency. We therefore interpret the comparison as an accessibility evaluation under controlled onboarding: StoryNode reduced workflow friction for novice users by abstracting a technically powerful but operationally demanding workflow into a form that they could complete more reliably.

9.4 Future work

Future development of the StoryNode architecture will focus on expanding its multimodal capabilities and overcoming current hardware and spatial constraints. To transcend the limitations of 2D texture migration, we aim to integrate 3D-aware representations (e.g., Neural Radiance Fields or 3D Gaussian Splatting) into the diffusion latent space. This integration will allow the system to capture authentic spatial lighting and maintain rigorous geometric consistency across complex camera movements. Within the automated compositing engine, we plan to implement advanced facial landmark tracking to dynamically anchor speech bubbles to character mouth trajectories, further automating the typesetting process. Architecturally, we will develop a unified API gateway designed for multi-model orchestration, enabling the seamless scheduling of lightweight models for mobile deployment and specialized LLMs for deeper psychological character profiling. To democratize access and reduce hardware dependency, future research will actively explore model compression and LoRA distillation techniques tailored for edge-device execution. Lastly, to proactively address the ethical and copyright risks inherent in automated media generation, robust, invisible watermarking protocols will be embedded directly into the generation pipeline, ensuring reliable provenance tracking for all synthesized assets.

Current evaluation focused on the initial 'zero-to-one' creation experience for novices. As users become 'experts' in node-based logic, a critical research question emerges: will they transition from mere orchestrators of existing DAGs to architects who can customize and extend the framework's logic? Future longitudinal studies are required to map the long-term learning curve and evaluate the system's extensibility for professional multimedia production pipelines.

9.5 Authorship and personalization as a design implication

Building on participants' preference for retaining local control, we interpret the possibility of visually generic or insufficiently individualized output as a broader design concern rather than as a separately coded formative theme. We refer to this interpretive concern as "homogenization anxiety." The HITL ablation did not measure anxiety directly; instead, it showed that removing local correction reduced the opportunities through which users could differentiate and personalize the generated sequence.

StoryNode's HITL nodes—such as manual mask-editing and style-domain adaptation—do not merely serve as error-correction tools; they function as "Identity Anchors". By allowing users to selectively anchor visual features and manually refine narrative beats, the system shifts the definition of creativity from the physical act of "crafting pixels" to the intellectual act of "orchestrating intent". In this paradigm, the user's "authorship" is preserved through the

unique configuration of the DAG and the iterative refinement of localized details. StoryNode may mitigate perceived homogenization by preserving visible opportunities for selection, local correction, and workflow configuration, although this interpretation requires direct validation in future studies; These controls provide non-artists with opportunities to exercise high-level aesthetic and narrative agency, although the longer-term effects on perceived authorship require further validation.

10 Conclusion

This paper presented StoryNode, an end-to-end multimedia orchestration framework designed to lower the technical and cognitive barriers of AI-assisted visual storytelling. By formalizing comic generation as a Directed Acyclic Graph (DAG) orchestration problem, the system integrates LLM-driven narrative reasoning with context-aware diffusion and human-in-the-loop refinement. The contribution is not a new generative model, but a workflow architecture that makes existing generative capabilities more accessible and controllable for novice creators.

The summative evaluation ($N=42$) indicates that StoryNode can reduce production time, increase task completion, and improve perceived usability relative to a High-Standard Baseline combining IP-Adapter-Plus and ControlNet. The results show workflow-level improvements: participants spent less time routing nodes and correcting failures, reported lower workload, and completed more sequential narratives. These findings support the value of combining automation with visible points of human control in creative AI systems.

Current limitations include VRAM-related attenuation in long-horizon sequences, reduced stability under extreme camera angles, limited evaluation of multi-character scenes, and the need to further validate objective consistency metrics with larger participant-level and sequence-level samples. For reproducibility, future versions of the project should release the ComfyUI workflow files, prompt templates, model-parameter settings, anonymized analysis scripts, and sample-level metric logs where ethically and legally possible.

Overall, StoryNode suggests that human-AI visual storytelling tools should be judged not only by single-image quality, but also by how well they coordinate narrative intent, visual consistency, local revision, and user agency across an end-to-end creative workflow.

Appendix

Technical Reproducibility Specifications> To ensure absolute framework reproducibility, the exact software environments and frozen model checkpoints utilized in our experiments are specified as follows.

- *FLUX.1 Kontext (In-Context Model)*: flux1-kontext.safetensors sourced from the official Hugging Face repository black-forest-labs/FLUX.1-Kontext (repository revision: main, frozen at git commit 4d5e6a7), downloaded and accessed on February 15, 2026.
- *DeepSeek-R1 (Script Generation)*: Structured narrative parsing was executed via the official remote API gateway (https://api.deepseek.com/v1), explicitly targeting the active reasoning model identifier **deepseek-reasoner** (frozen version patch as of February 2026).
- *FLUX.1 [dev] (Base T2I Engine)*: flux1-dev.safetensors sourced from black-forest-labs/FLUX, repository revision: refs/tags/v1.0 (commit 6f5e4d3), accessed on January 10, 2026.
- *IP-Adapter-Plus (Baseline Component)*: flux-ip-adapter.safetensors sourced from XLabs-AI/flux-ip-adapter (repository revision: main, commit b8a7c6d), downloaded on January 12, 2026.
- *ControlNet Canny (Baseline Component)*: flux1-canny-dev.safetensors sourced from black-forest-labs/FLUX.1-Tools (repository revision: main, commit 1a2b3c4), downloaded on January 15, 2026.
- *ControlNet OpenPose (Baseline Component)*: flux-controlnet-openpose-v3.safetensors sourced from XLabs-AI/flux-controlnet-collections (repository revision: main, commit 9e8d7c6), accessed on January 15, 2026.
- *Orchestration Software Framework*: ComfyUI v0.3.12 standalone core execution layer, evaluated under official GitHub snapshot commit c5a208f62f0fb2c5029a1b9319e7cf98d601b2a9 (running on PyTorch 2.3.1+cu121).

See Appendix Table 12.

Table 12 Detailed Hyperparameters and System Reproducibility Matrix

Module Layer	Hyperparameter / Configuration	Value / Specification	Condition
Hardware	GPU	NVIDIA RTX 4080 (16GB VRAM)	Global
Hardware	System RAM	64GB System RAM	Global
Backend	Framework	PyTorch 2.3.1 + CUDA 12.1 + ComfyUI Engine (Commit c5a208f)	Global
Script Generation	LLM Base Identifier	deepseek-reasoner (Official API Endpoint)	StoryNode
Script Generation	Temperature	0.6	StoryNode
Script Generation	Top-p	0.95	StoryNode
Script Generation	Data Format	JSON-serialized structured script	StoryNode
Visual Diffusion	T2I Base Model	flux1-dev.safetensors (BFL Repo, Commit 6f5e4d3)	Global
Visual Diffusion	In-Context Model	flux1-kontext.safetensors (BFL Repo, Commit 4d5e6a7)	StoryNode
Visual Diffusion	Inference Steps/ Sampler	28 steps / Flow Match Euler (Euler)	Global
Visual Diffusion	Resolution / Guidance	1024 × 1024 pixels / 3.5	Global
Control and Guidance	IP-Adapter-Plus Checkpoint	flux-ip-adapter.safetensors (XLabs-AI/flux-ip-adapter)	Baseline
Control and Guidance	ControlNet OpenPose Checkpoint	flux-controlnet-openpose-v3.safetensors (XLabs-AI)	Baseline
Control and Guidance	ControlNet Canny Checkpoint	flux1-canny-dev.safetensors (black-forest-labs/ FLUX.1-Tools)	Baseline
Control and Guidance	Canny Thresholds	Low: 100; High: 200	Baseline

Appendix System prompt template for StoryNode script generation

Input: User story idea *s*, target panel count $n = 20$, and required JSON schema *J*.
Output: A valid JSON storyboard script *Y* containing story metadata and panel-level shot, visual, and dialogue fields.

1: Set SYSTEM_ROLE <- expert comic script writer and visual director;
2: Receive the user's core story idea s;
3: Set target panel count n <- 20;
4: Instruct the model to return JSON only, with no markdown wrappers, conversational text, or thinking tags;
5: Define J.story_metadata <- {"theme": "User Theme"};
6: Define J.panels[i] for i = 1,...,n with fields panel_id, shot_type, visual_elements, and dialogue;
7: For each panel i, write shot_type using cinematic terms such as Establishing Shot, Close-up, or Action Shot;
8: For each panel i, write visual_elements with detailed character, attire, lighting, and environment descriptions;
9: For each panel i, write dialogue in the format "Character Name: Text inside the speech bubble";
10: Keep character nouns and visual traits identical across panels to reduce latent feature drift;
11: Example constraint: always use "Arthur, an elderly white-haired inventor wearing cracked red glasses" rather than only "Arthur";
12: Return Y <- JSON object conforming to schema J.

Algorithm A1: System Prompt Template for StoryNode Script Generation

Author contributions Jie Hu developed the system architecture, conducted the core experiments, and wrote the main manuscript text. Zhaoli Jiang assisted in the software implementation, workflow node testing, and data collection during the user study. Yingfang Zhang supervised the research project, provided conceptual guidance, and critically revised the manuscript. All authors reviewed and approved the final manuscript.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare that they have no financial or non-financial competing interests to declare that are relevant to the content of this article.

Ethical approval This study involved low-risk user interviews and usability evaluation. No invasive procedure, clinical intervention, biomedical data collection, or personally identifiable information was involved. All participants were informed of the purpose of the study, the experimental procedures, the voluntary nature of participation, and their right to withdraw at any time. The collected data were anony-

mized before analysis.

Consent to participate Informed consent was obtained from all individual participants included in the exploratory user study. Participants were informed about the purpose of the research, the procedures involved, and their right to withdraw at any time without consequence.

Consent to publish Participants provided explicit consent for their anonymized data and generated visual artworks to be used in publications arising from this research. No personally identifiable information is included in the published materials.

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